T1 · Probability & concentration

Q1 · Error reduction for decision problems

Theme of MIT 6.889 HW0 · difficulty ★ / 考点参考 MIT 6.889 HW0 · 难度 ★

A randomized algorithm A for a decision problem errs with probability at most $1/4$ on every input and runs in time T. Design B that errs with probability at most β and runs in time $O(T \log(1/\beta))$. Prove it.

判定问题的随机算法 A 在每个输入上出错概率至多 $1/4$，运行时间 T。设计出错概率至多 β、时间 $O(T \log(1/\beta))$ 的算法 B，并证明。

Q2 · Coupon collector

Theme of MIT 6.889 HW0 and L1 balls-into-bins · ★ / 考点参考 MIT 6.889 HW0 与 L1 球箱模型 · ★

A fair n-sided die is rolled repeatedly. (a) Show that the expected number of rolls until every face has appeared is $nH_n = n \ln n + O(n)$. (b) Show that after $2n \ln n$ rolls, every face has appeared with probability at least $1 - 1/n$.

反复掷一个均匀的 n 面骰子。(a) 证明所有面都出现所需的期望次数为 $nH_n = n \ln n + O(n)$。(b) 证明掷 $2n \ln n$ 次后，所有面都出现的概率至少 $1 - 1/n$。

Q3 · Average number of hashtags

Theme of NUS CS5234 (2019) Tutorial 4 · ★★ / 考点参考 NUS CS5234（2019）Tutorial 4 · ★★

A stream of tweets arrives; each tweet contains between 0 and 70 hashtags. Using a uniform sample of s tweets drawn with replacement, estimate the average number A of hashtags per tweet within additive error ε, with probability at least $2/3$. How large must s be? What changes if you want error εA instead and know $A \geq 1$?

推文流中每条推文含 0 到 70 个话题标签。用有放回均匀抽取的 s 条推文，以至少 $2/3$ 的概率把平均标签数 A 估计到加性误差 ε 以内。s 需要多大？若要求误差为 εA 且已知 $A \geq 1$，有何变化？

T2 · Query algorithms & lower bounds

Q4 · Boosting an approximation scheme

Theme of MIT 6.889 HW0 · ★ / 考点参考 MIT 6.889 HW0 · ★

Algorithm A outputs a $(1+\varepsilon)$-approximation of a positive quantity V with probability $3/4$. Build B that outputs a $(1+\varepsilon)$-approximation with probability $1-\delta$, using $O(\log(1/\delta))$ calls to A. Why does taking the average of the runs not work?

算法 A 以 $3/4$ 的概率输出正数 V 的 $(1+\varepsilon)$ 近似。构造只调用 $O(\log(1/\delta))$ 次 A、以 $1-\delta$ 概率输出 $(1+\varepsilon)$ 近似的算法 B。为什么取平均不行？

Q5 · Element distinctness needs $\Omega(n)$

Reduction pattern of L3–L4 · ★ / L3–L4 的归约套路 · ★

Given query access to $a_1, \ldots, a_n$ (n even), decide exactly whether all values are distinct. Prove that the randomized query complexity is $\Omega(n)$ by a reduction from OR.

可查询 $a_1, \ldots, a_n$（n 为偶数），精确判断所有值是否互不相同。用从 OR 的归约证明随机查询复杂度为 $\Omega(n)$。

Q6 · Finding the maximum needs $\Omega(n)$

Yao's principle for a search problem · ★★ / Yao 原理用于搜索问题 · ★★

Given query access to $a_1, \ldots, a_n$, output an index of a maximum entry, correct with probability $\geq 2/3$. Prove that $\Omega(n)$ queries are needed. Hint: the answer is not binary, but Yao still applies.

可查询 $a_1, \ldots, a_n$，输出一个最大值所在的下标，正确率 $\geq 2/3$。证明需要 $\Omega(n)$ 次查询。提示：答案不是二元的，但 Yao 原理仍然适用。

Q7 · Average degree with additive error

Theme of L2 · ★ / L2 考点 · ★

In the adjacency-list model (degree queries), estimate the average degree $\bar{d}$ of an n-vertex graph within additive error εn with probability $2/3$. How many queries suffice? Why does the same idea fail for a $(1 \pm \varepsilon)$-approximation of $\bar{d}$ on sparse graphs?

在邻接表模型（可查询度数）中，以 $2/3$ 的概率把 n 个顶点的图的平均度 $\bar{d}$ 估计到加性误差 εn 以内。需要多少次查询？为什么同样的做法在稀疏图上做不到 $(1 \pm \varepsilon)$ 近似？

T3 · Property testing

Q8 · Testing palindromes

Generic "far ⇒ many witnesses" pattern · ★ / 「far ⇒ 大量证据」通用套路 · ★

A string $x \in \{0,1\}^n$ is a palindrome if $x_i = x_{n+1-i}$ for all i. (a) What is the distance of x from being a palindrome? (b) Give a one-sided tester with $O(1/\varepsilon)$ queries.

若对所有 i 有 $x_i = x_{n+1-i}$，称 $x \in \{0,1\}^n$ 为回文串。(a) x 到回文串的距离是多少？(b) 给出 $O(1/\varepsilon)$ 次查询的单边测试器。

Q9 · Testing connectivity in bounded-degree graphs

Theme of MIT 6.889 HW1 (Goldreich–Ron) · ★★★ / 考点参考 MIT 6.889 HW1（Goldreich–Ron） · ★★★

A graph has maximum degree d and is given by adjacency lists. It is ε-far from connected if more than εdn edge modifications are needed to make it connected. (a) Show that an ε-far graph has more than εdn connected components (you may assume that adding one edge can always merge two components while keeping the degree bound, possibly after removing one edge). (b) Deduce that more than $\frac{\varepsilon dn}{2}$ components have at most $\frac{2}{\varepsilon d}$ vertices. (c) Give a one-sided tester whose query complexity is independent of n.

图的最大度为 d，以邻接表给出。若需要超过 εdn 次改边才能连通，称其 ε-far。(a) 证明 ε-far 的图有超过 εdn 个连通分量（可以假设总能用一次加边合并两个分量并保持度数上界，必要时先删一条边）。(b) 推出其中超过 $\frac{\varepsilon dn}{2}$ 个分量的顶点数不超过 $\frac{2}{\varepsilon d}$。(c) 给出查询复杂度与 n 无关的单边测试器。

Q10 · Distance between distributions

Theme of L4 and MIT 6.889 HW3 · ★★ / 考点参考 L4 与 MIT 6.889 HW3 · ★★

(a) Show $\Delta_{\mathsf{TV}}(\mu,\nu) = \max_{A \subseteq [n]} |\mu(A) - \nu(A)|$. (b) Let μ be uniform on $[n]$ and ν uniform on a fixed set S of size $n/4$. Compute $\Delta_{\mathsf{TV}}(\mu,\nu)$ and $\|\nu\|_2^2$. (c) How many samples does the collision tester of L4 need to tell them apart, up to constants?

(a) 证明 $\Delta_{\mathsf{TV}}(\mu,\nu) = \max_{A \subseteq [n]} |\mu(A) - \nu(A)|$。(b) μ 为 $[n]$ 上的均匀分布，ν 为某个固定集合 S（$|S| = n/4$）上的均匀分布。计算 $\Delta_{\mathsf{TV}}(\mu,\nu)$ 与 $\|\nu\|_2^2$。(c) L4 的碰撞测试器大约需要多少样本才能区分二者？

T4 · Streaming

Q11 · Morris++

Theme of NUS CS5234 (2019) Tutorial 4 · ★★ / 考点参考 NUS CS5234（2019）Tutorial 4 · ★★

Using the Morris counter (unbiased, Var $< A^2$), design a streaming algorithm that outputs $\hat{A} \in [(1-\varepsilon)A, (1+\varepsilon)A]$ with probability $1-\delta$. What is its space?

利用 Morris 计数器（无偏，Var $< A^2$），设计一个以 $1-\delta$ 概率输出 $\hat{A} \in [(1-\varepsilon)A, (1+\varepsilon)A]$ 的流式算法。空间是多少?

Q12 · Weighted reservoir sampling

Classic variant of L5 reservoir sampling · ★★ / L5 蓄水池抽样的经典变形 · ★★

Items $x_1, x_2, \ldots$ arrive with positive weights $w_1, w_2, \ldots$. Keep one item so that at the end x_i is kept with probability w_i / W_n, where $W_i = \sum_{j \le i} w_j$. Give the algorithm and prove it.

元素 $x_1, x_2, \ldots$ 依次到达，带正权重 $w_1, w_2, \ldots$。只保留一个元素，使结束时 x_i 被保留的概率为 w_i / W_n，其中 $W_i = \sum_{j \le i} w_j$。给出算法并证明。

Q13 · Cycles and spanners in a stream

Theme of L5 · ★★ / L5 考点 · ★★

Edges of an n-vertex graph arrive in a stream. (a) Decide whether the graph contains a cycle using $O(n \log n)$ bits. (b) Run the greedy spanner with $k = 2$. Show that the output H has no cycle of length ≤ 4 and that $d_H(u,v) \le 3\, d_G(u,v)$. (c) Why is $|E(H)| = O(n^{3/2})$?

n 个顶点的图的边以流的形式到达。(a) 用 $O(n \log n)$ 位判断图中是否有环。(b) 用 $k = 2$ 的贪心 spanner，证明输出 H 没有长度 ≤ 4 的环，且 $d_H(u,v) \le 3\, d_G(u,v)$。(c) 为什么 $|E(H)| = O(n^{3/2})$?